Source: Sahil Bhola, Self-contained Proof for a Mixed-precision Block Jacobi Algorithm.

Selected deliverable: the full active result labeled
theorem:mixed_precision_update_form_block_jacobi_in_floating_point_arithmetic in main.tex (lines 703-1270).

The original mathematical proof is available in paper.tex. The selected statement below is authoritative.
Earlier statements and notation supply context. A referenced result may be proved as a helper. Do not assume a
desired algorithmic bound or an unproved paper lemma as an added hypothesis.

SELECTED STATEMENT (verbatim LaTeX, proof omitted)

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\begin{theorem}[Mixed-precision Update-form Block Jacobi in Floating-point
Arithmetic]\label{theorem:mixed_precision_update_form_block_jacobi_in_floating_point_arithmetic}
Let $\mat{a}\ireal{n\times n}$ be decomposed as
$\mat{a}=\mat{d}+\mat{n}$, where $\mat{d}$ is block diagonal with
symmetric positive-definite blocks of size $T\times T$, and let
$\vect{x}$ satisfy $\mat{a}\vect{x}=\vect{b}$. Consider the update-form
block-Jacobi iteration in which the matrix--vector product, residual, and
solution update are computed in precision $\urd_s$, while each correction
system is solved by Cholesky factorization in precision $\urd_c$, with
$\urd_s<\urd_c$. Assume that $S$ divides $n$, that all relevant
$\gamma$-constants are defined, and that no overflow or underflow occurs.

Define
\begin{align}
    \gamma_{m;s}
    &\define \frac{m\urd_s}{1-m\urd_s},
    &
    \gamma_{m;c}
    &\define \frac{m\urd_c}{1-m\urd_c},
    \\
    c_1
    &\define
    \gamma_{S+n/S-1;s}
    +\urd_s
    +\urd_s\gamma_{S+n/S-1;s},
    &
    c_2
    &\define \gamma_{T;c}^2+3\gamma_{T+1;c}.
\end{align}
Let $\widehat{\mat{r}}$ be block diagonal, with each diagonal block equal
to the computed Cholesky factor of the corresponding block of $\mat{d}$,
and define
\begin{align}
    \mat{g}
    &\define
    c_2\abs{\mat{d}^{-1}}
    \abs{\widehat{\mat{r}}^T}
    \abs{\widehat{\mat{r}}},
    &
    \eta
    &\define \norm[\infty]{\mat{g}},
    \\
    q
    &\define \norm[\infty]{\mat{d}^{-1}\mat{n}},
    &
    p
    &\define
    \norm[\infty]{\abs{\mat{d}^{-1}}\abs{\mat{a}}}.
\end{align}
If $\eta<1$, then the computed iterates
$\hat{\vect{x}}^k$ and errors
$\vect{e}^k\define\hat{\vect{x}}^k-\vect{x}$ satisfy
\begin{align}
    \norm[\infty]{\vect{e}^{k+1}}
    \leq
    \alpha\norm[\infty]{\vect{e}^k}+\beta,
\end{align}
where
\begin{align}
    \alpha
    &\define
    \frac{
        \eta+\urd_s+(1+\urd_s)(q+\urd_c p)
    }{1-\eta}
    +\urd_s
    +\frac{(1+\urd_s)(1+\urd_c)c_1p}{1-\eta},
    \\
    \beta
    &\define
    \left[
        \urd_s
        +\frac{(1+\urd_s)(1+\urd_c)c_1p}{1-\eta}
    \right]
    \norm[\infty]{\vect{x}}
    \\
    &\qquad
    +\frac{(1+\urd_s)(1+\urd_c)\urd_s}{1-\eta}
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     \norm[\infty]{\mat{d}^{-1}}
     \norm[\infty]{\vect{b}}.
\end{align}
Consequently, if $\alpha<1$, then for every $k\geq0$,
\begin{align}
    \norm[\infty]{\vect{e}^{k}}
    \leq
    \alpha^k\norm[\infty]{\vect{e}^{0}}
    +\frac{1-\alpha^k}{1-\alpha}\beta,
\end{align}
and
\begin{align}
    \limsup_{k\to\infty}
    \norm[\infty]{\vect{e}^{k}}
    \leq \frac{\beta}{1-\alpha}.
\end{align}
\end{theorem}

SECTION CONTEXT
\section{Block Jacobi Error Analysis in Floating-point Arithmetic (Mixed-precision)}
A key observation is that the correction term $\vect{c}^{k+1}\to 0$ (In Floating) as the solution improves.
Thus, for large $k$, we do not even need to be very accurate in computing the correction term, and therefore, we
can use low-precision arithmetic ($\urd_c$) compared to the precision in which the solution is computed $\urd_s
< \urd_c$.

EARLIER STATEMENTS FOR CONTEXT
\begin{theorem}[Block Matrix-vector Product]
\label{theorem:block_matrix_vector_product}
Let $\mat{a}\ireal{m\times n}$ and $\vect{x}\ireal{n}$, and suppose that
each component of $\vect{y}=\mat{a}\vect{x}$ is evaluated using block dot
products of size $T$. Within each block, the products are accumulated
sequentially, and the resulting
$n^\dagger=\ceil{n/T}$ partial sums are also reduced sequentially. Entries
beyond $n$ in the final block are taken to be zero. Assume that no
overflow or underflow occurs and that
$(T+n^\dagger-1)\urd<1$. Then the computed result $\hat{\vect{y}}$
satisfies
\begin{align}
    \hat{\vect{y}}
    &=(\mat{a}+\Delta\mat{a})\vect{x},
    &
    \abs{\Delta\mat{a}}
    &\leq \gamma_{T+n^\dagger-1}\abs{\mat{a}},
\end{align}
where the inequality is interpreted componentwise and
\begin{align}
    \gamma_{T+n^\dagger-1}
    \define
    \frac{(T+n^\dagger-1)\urd}
         {1-(T+n^\dagger-1)\urd}.
\end{align}
\end{theorem}

\begin{theorem}[Cholesky Solve]\label{theorem:cholesky_solve}
Let $\mat{a}\ireal{n\times n}$ be symmetric positive definite and let
$\vect{b}\ireal{n}$. Suppose that $\mat{a}\vect{x}=\vect{b}$ is solved in
floating-point arithmetic by computing an upper-triangular Cholesky factor
$\widehat{\mat{r}}$, followed by forward and backward substitution. Assume
that the factorization completes successfully, no overflow or underflow
occurs, and $(n+1)\urd<1$. Then the computed solution
$\hat{\vect{x}}$ satisfies
\begin{align}
    (\mat{a}+\Delta\mat{a})\hat{\vect{x}}=\vect{b},
\end{align}
where
\begin{align}
    \abs{\Delta\mat{a}}
    &\leq
    \left(\gamma_{n+1}+2\gamma_n+\gamma_n^2\right)
    \abs{\widehat{\mat{r}}^T}\abs{\widehat{\mat{r}}}
    \\
    &\leq
    \left(3\gamma_{n+1}+\gamma_n^2\right)
    \abs{\widehat{\mat{r}}^T}\abs{\widehat{\mat{r}}}.
\end{align}
The inequalities are interpreted componentwise.
\end{theorem}

\begin{theorem}[Update-form Block Jacobi in Real Arithmetic]
\label{theorem:block_jacobi_real}
Let $\mat{a}\ireal{n\times n}$ be decomposed as
$\mat{a}=\mat{d}+\mat{n}$, where $\mat{d}$ is a nonsingular block-diagonal
matrix and $\mat{n}$ contains the off-diagonal blocks. Let
$\vect{x}$ satisfy $\mat{a}\vect{x}=\vect{b}$, and consider the exact
block-Jacobi iteration
\begin{align}
    \mat{d}\vect{c}^{k+1}
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    &=\vect{b}-\mat{a}\vect{x}^k,
    &
    \vect{x}^{k+1}
    &=\vect{x}^k+\vect{c}^{k+1}.
\end{align}
Define the forward error $\vect{e}^k\define\vect{x}^k-\vect{x}$ and the
iteration matrix $\mat{p}\define-\mat{d}^{-1}\mat{n}$. Then
\begin{align}
    \vect{e}^{k+1}=\mat{p}\vect{e}^k,
    \qquad
    \vect{e}^k=\mat{p}^k\vect{e}^0,
\end{align}
and hence
\begin{align}
    \abs{\vect{e}^k}
    &\leq \abs{\mat{p}}^k\abs{\vect{e}^0},
    &
    \norm[\infty]{\vect{e}^k}
    &\leq
    \norm[\infty]{\mat{p}}^k
    \norm[\infty]{\vect{e}^0}.
\end{align}
The iteration converges to $\vect{x}$ for every initial guess if and only
if
\begin{align}
    \rho(\mat{p})
    =\rho(\mat{d}^{-1}\mat{n})<1.
\end{align}
In particular, $\norm[\infty]{\mat{p}}<1$ is a sufficient condition for
convergence and makes the iteration a contraction in the infinity norm.
\end{theorem}

COMMON ROUNDING MODEL AND ERROR CONSTANTS
\section{Floating-Point Model}
We assume the standard floating-point model~\cite{higham2002accuracy}: for every elementary
operation $\circ \in \{+,-,\times,/\}$,
\begin{equation}
    \fl(a \circ b) = (a \circ b)(1+\delta),
    \qquad
    \abs{\delta} \le \urd,
    \label{eq:fp-model}
\end{equation}
where $\urd$ is the unit roundoff and $\delta$ is the relative roundoff error.
Using~\cref{eq:fp-model} for numerical analysis we typicall obtain the product
\begin{align}
    \prod_{i=1}^n (1 + \delta_i)^{\rho_i} \define 1 + \theta_n, \qquad \rho = \pm 1,
    \label{eq:theta_n}
\end{align}
where $n$ denotes the number of accumulated roundoff error terms.
For numerical analysis, we seek reliable and sharp bounds for $\theta_n$.


\section{Deterministic Rounding Error Analysis}
The deterministic rounding error analysis proposes the following problem-size dependent constant
\begin{align}
\gamma_{n} \define \frac{n\urd}{1 - n\urd},
\label{eq:drea}
\end{align}
for $n\urd < 1$.
Here, $n$ is the number of accumulated error terms~\cite{higham2002accuracy}.
The problem-size dependent constant is monotonic, that is, $\gamma_{n+1} > \gamma_{n}$ for $\urd > 0$.
Performing backward/forward error analysis using~\cref{eq:drea} we obtain
\begin{align}
    \abs{\theta_n} \leq \gamma_n.
\end{align}

SOURCE NOTATION MACROS
\newcommand{\urd}{\ensuremath{\mathrm{u}}}
\newcommand{\define}{\ensuremath{\triangleq}}
\newcommand{\vect}[1]{\mathbf{\MakeLowercase{#1}}}
\newcommand{\mat}[1]{\mathbf{\MakeUppercase{#1}}}
\newcommand{\ireal}[1]{\in\mathbb{R}^{#1}}
\newcommand{\norm}[2][]{\left\|#2\right\|_{#1}}
\DeclareMathOperator{\fl}{fl}
\newcommand{\R}{\mathbb{R}}
\newcommand{\abs}[1]{\left|#1\right|}
\newcommand{\ceil}[1]{\left\lceil #1 \right\rceil}
\newcommand{\widehatgamma}[1]{\widehat{\gamma}_{#1}}
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